

MINERAL QUALITY & FEASIBILITY ANALYZER

Prototype 2 — Professional Project Documentation

Python-based screening and decision-support application for mineral quality and basic economic feasibility.

Item	Details
Project Type	Independent technical/software prototype
Domain	Mining, Mineral Processing & Decision Support
Programming Language	Python
Libraries	Pandas, Matplotlib
Application Interface	Streamlit web application
Main File	mineral quality prototype2.py

1. Executive Summary

The Mineral Quality & Feasibility Analyzer is a software prototype that provides a structured preliminary assessment of a mineral or ore sample using chemical, physical, behavioural and economic inputs. The system converts selected inputs into normalized scores, combines them using predefined weights, calculates an overall Mineral Quality Score out of 100, estimates margin per tonne and generates a visual report.

Prototype 2 is designed as a transparent decision-support and interview/portfolio project. It is not a certified laboratory measurement system and should not be presented as a replacement for laboratory assays, metallurgical testing or formal feasibility studies.

2. Problem Statement

Mineral evaluation involves several variables rather than a single measurement. Chemical composition, stability, reactivity, material consistency, moisture and economic conditions can all affect preliminary suitability. The project creates a repeatable digital workflow so these inputs can be evaluated consistently and reported in one place.

3. Objectives

- Create a structured digital workflow for preliminary mineral/ore assessment.
- Normalize selected mineral parameters into comparable 0–100 scores.
- Combine selected indicators into an overall quality score.
- Calculate basic economic margin using price and cost per tonne.
- Classify quality and economic condition.
- Interpret pH using a simple descriptive classification.
- Visualize the individual quality dimensions.
- Save a structured CSV report.
- Provide a foundation for later laboratory, sensor, database and validated-model integration.

4. Scope

Current prototype includes:

- Mineral/ore name and sample ID.
- Valuable component and impurity percentages.
- pH and pH classification.
- Chemical stability and reactivity ratings.
- Particle size, bulk density and hardness.
- Moisture and homogeneity.
- Market price and mining, processing, transport and other costs.
- Weighted quality scoring, economic margin, classification, chart and CSV report.

Outside the current scope:

- Direct laboratory/sensor measurement.
- Automatic mineral identification.
- Geological reserve estimation.
- Detailed metallurgical recovery simulation.

- Full bankable feasibility analysis.
- Machine-learning prediction from a validated industrial dataset.

5. System Workflow

Stage	Function	Output
1. Identification	Capture mineral and sample information	Sample record
2. Chemical	Capture composition, impurities and pH	Chemical indicators
3. Behaviour	Capture stability and reactivity ratings	Behaviour scores
4. Physical	Capture particle size, density, hardness and moisture	Physical indicators
5. Homogeneity	Capture consistency rating	Homogeneity score
6. Economics	Calculate total cost and margin	Economic result
7. Scoring	Apply normalization and weights	Quality score
8. Reporting	Display chart and save CSV	Analysis report

6. Input Parameters

Category	Parameter	Unit / Type	Purpose
Identification	Mineral / Ore name	Text	Identifies material
Identification	Sample ID	Text	Tracks sample
Chemical	Main valuable component	Text	Names target component
Chemical	Main component	%	Valuable fraction
Chemical	Total impurities	%	Non-target fraction
Chemical	pH	Numeric	Acidity/alkalinity
Behaviour	Chemical stability	1–5	Stability assessment
Behaviour	Reactivity	1–5	Unwanted reactivity assessment
Physical	Particle size	mm	Particle scale
Physical	Bulk density	g/cm ³	Bulk material density
Physical	Hardness	Mohs	Relative hardness
Physical	Moisture	%	Moisture content
Physical	Homogeneity	1–5	Material consistency
Economic	Market price	Currency/tonne	Expected selling value
Economic	Mining cost	Currency/tonne	Extraction cost
Economic	Processing cost	Currency/tonne	Processing cost
Economic	Transportation cost	Currency/tonne	Transport cost
Economic	Other cost	Currency/tonne	Additional cost

7. Calculation Methodology

7.1 Composition Score

$$\text{Composition Score} = [\text{Main Component \%} / (\text{Main Component \%} + \text{Impurity \%})] \times 100$$

This is a simplified composition indicator and should be adapted when multiple valuable minerals or impurity classes are involved.

7.2 Stability Score

$$\text{Stability Score} = (\text{Stability Rating} / 5) \times 100$$

7.3 Reactivity Score

$$\text{Reactivity Score} = [(6 - \text{Reactivity Rating}) / 5] \times 100$$

The prototype interprets the defined reactivity rating so that lower unwanted reactivity contributes to a higher screening score.

7.4 Homogeneity Score

$$\text{Homogeneity Score} = (\text{Homogeneity Rating} / 5) \times 100$$

7.5 Moisture Score

Moisture	Score
≤ 5%	100
>5% to 10%	80
>10% to 15%	60
>15% to 20%	40
>20%	20

These thresholds are prototype screening assumptions; actual limits depend on mineral, process and product specifications.

7.6 Economic Calculation

$$\text{Total Cost} = \text{Mining} + \text{Processing} + \text{Transportation} + \text{Other Cost}$$

$$\text{Estimated Margin} = \text{Market Price} - \text{Total Cost}$$

The current prototype assigns an economic score of 100 for positive margin and 0 for non-positive margin, while zero margin is classified separately as break-even.

7.7 Overall Quality Score

$$\text{Quality Score} = 0.25C + 0.15S + 0.15R + 0.15H + 0.10M + 0.20E$$

C = composition, S = stability, R = reactivity, H = homogeneity, M = moisture and E = economics.

Parameter	Weight
Composition	25%
Stability	15%
Reactivity	15%
Homogeneity	15%
Moisture	10%
Economics	20%

7.8 Quality Classification

Quality Score	Class
≥ 85	HIGH QUALITY
70–84.99	GOOD QUALITY
50–69.99	MODERATE QUALITY
< 50	LOW QUALITY

7.9 pH Interpretation

pH	Classification
< 4	Strongly acidic
4 to < 6	Acidic
6 to 8	Approximately neutral
> 8 to 10	Alkaline
> 10	Strongly alkaline

8. Software Architecture

The application follows an input → calculation → classification → visualization → reporting workflow.

- Input layer: collects mineral, laboratory/assessment and economic values.
- Validation layer: constrains rating values and converts numeric entries.
- Calculation layer: derives normalized scores and economic margin.
- Decision layer: assigns pH, economic and quality classifications.
- Visualization layer: displays a bar chart of quality dimensions.
- Reporting layer: exports the final record to mineral_quality_report.csv.

9. Technology Stack

Technology	Purpose
Python	Core programming and calculation logic
Pandas	Tabular data handling and CSV report generation
Matplotlib	Scientific visualization
Streamlit	Interactive browser-based application interface
CSV	Portable report format

10. Outputs

- Mineral and sample identification.
- Composition, stability, reactivity, homogeneity and moisture scores.
- pH classification.
- Market price, total cost and estimated margin per tonne.
- Economic status: Economically Feasible, Break-Even or Not Economically Feasible.
- Overall Mineral Quality Score out of 100.
- Quality class.
- Quality comparison bar chart.
- CSV report: mineral_quality_report.csv.

11. Validation Considerations

The calculations are deterministic, so identical inputs produce identical outputs. However, the numerical assumptions are not automatically validated merely because the calculations are reproducible.

- Calibrate weights against historical laboratory and plant data.
- Use mineral-specific moisture and quality thresholds.
- Validate composition logic against actual assay methods.
- Expand economic modelling to include recovery, yield, energy, labour, royalties, depreciation and other relevant costs.
- Compare outputs against expert geological/metallurgical assessment before operational use.

12. Limitations

- Current weights are predefined prototype assumptions.
- Current output is an estimated screening score, not a direct physical measurement.
- Particle size, density and hardness are collected but are not currently part of the weighted quality formula.

- Economic scoring is simplified to a positive/non-positive margin rule.
- No uncertainty/confidence interval is produced.
- No direct laboratory, sensor or database integration is included.
- Not intended to replace certified assays or formal feasibility studies.

13. Future Scope

- Build a polished Streamlit dashboard.
- Add robust input validation and unit handling.
- Store multiple samples in a database.
- Add historical trends and sample comparison.
- Integrate laboratory assay data.
- Create mineral-specific scoring profiles.
- Calibrate or train predictive models using validated historical datasets.
- Add uncertainty and sensitivity analysis.
- Integrate appropriate plant sensors.
- Export professional PDF/Excel reports.
- Develop a more complete economic model.

14. Interview / Placement Explanation

Suggested 45–60 second explanation:

I developed a Python-based Mineral Quality & Feasibility Analyzer as a decision-support prototype. It accepts chemical, physical, behavioural and economic parameters for a mineral sample, converts selected parameters into normalized 0–100 scores, combines them using predefined weights, calculates an overall quality score and estimates basic economic margin per tonne. I used Pandas for structured report generation, Matplotlib for visualization and Streamlit to turn the Python workflow into an interactive web application. The current version is rule-based and transparent; my next step would be to validate and calibrate the scoring system using real laboratory and plant data, and then evaluate whether a machine-learning model is justified.

15. Common Interview Questions

Question	Suggested Answer
Is this machine learning?	The current Prototype 2 scoring engine is rule-based. A validated historical dataset could be used later for machine-learning prediction.
How is the quality score calculated?	Selected normalized scores are combined using the predefined weights: composition 25%, stability 15%, reactivity 15%, homogeneity 15%, moisture 10% and economics 20%.
Why use weights?	They provide a transparent way to represent the intended relative contribution of different indicators and can later be calibrated using domain or historical data.
How is economic feasibility calculated?	Total cost is the sum of mining, processing, transportation and other cost per tonne; margin is selling price minus total cost.
Does it physically measure mineral quality?	No. It calculates an estimated screening score from supplied inputs. Physical measurement requires suitable laboratory or field instruments.
What is the main limitation?	The scoring rules and weights are generic prototype

	assumptions and must be validated for a specific mineral and process.
How would you improve it?	I would integrate validated assays, mineral-specific specifications, database storage, sensitivity analysis and, where appropriate, machine-learning models.

16. Resume / CV Description

Mineral Quality & Feasibility Analyzer | Python, Pandas, Matplotlib, Streamlit

- Developed a Python-based decision-support prototype for preliminary mineral/ore quality and economic screening.
- Implemented transparent normalized scoring for composition, stability, reactivity, homogeneity and moisture.
- Calculated total cost and estimated margin per tonne for basic economic feasibility screening.
- Automated quality classification, pH interpretation and graphical comparison of quality parameters.
- Implemented CSV-based structured reporting.
- Designed the workflow for conversion into an interactive Streamlit web application.

17. Testing Checklist

Test Case	Expected Result
Valid sample input	Analysis completes and report is generated
Stability outside 1–5	Rating is constrained to 1–5
Reactivity outside 1–5	Rating is constrained to 1–5
Positive margin	Economically Feasible
Zero margin	Break-Even
Negative margin	Not Economically Feasible
Score ≥85	HIGH QUALITY
Score 70–84.99	GOOD QUALITY
Score 50–69.99	MODERATE QUALITY
Score <50	LOW QUALITY
CSV export	mineral_quality_report.csv created
Graph	Six scoring dimensions displayed

18. Conclusion

Prototype 2 demonstrates how a multidisciplinary mineral-evaluation workflow can be organized into a repeatable software process. Its key strengths are transparent calculations, structured reporting and the ability to combine technical and basic economic indicators in one workflow. For professional deployment, the most important next step is validation: the weights, thresholds and scoring logic should be calibrated against mineral-specific laboratory, plant and commercial data before the application is used for operational or investment decisions.

Appendix — Recommended Application Structure

```
mineral quality prototype 2/
├─ mineral quality prototype2.py
├─ mineral_quality_report.csv
└─ start_mining_quality_prototype2.bat
```

Technical note: The score should be described as an estimated screening score, not as a universal mineral-quality standard. This wording is important in interviews because it demonstrates awareness of model validation and engineering limitations.